pfi

Python for Fixed Income

COURSE 7 · NLP, RAG, AND LLMS

First LLM Request

WEEK 3 · DAY 1

Created by **Jeff Lenamon**

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Today: first LLM requests

Two documents

FOMC	the statement of 2026-09-16
Indenture	Boyd, Section 3.07(b), one paragraph
Saved	responses recorded 2026-10-08

Today

1. A request, and OpenRouter
2. The key, kept out of Git
3. ask: one door for every request
4. First request, checked by regex
5. Tokens, cost, reasoning off
6. Temperature; replay; failures
7. OpenAI, Excel, Git, AI check

Every response in the notebook is a saved response, replayed. A live run can differ.

The rules for real documents

Every prompt asks what a document says: a quote, a summary, or NOT IN TEXT

Allowed names	Never called
the response	the model's opinion or view
the saved response	what the model believes
the model returned	what the model knows

The system prompt goes first in every request and refuses an opinion, a grade, a comparison of issuers, or a view about the future.

What a request is

```
{
  "model": "nvidia/nemotron-3-super-120b-a12b:free",
  "messages": [
    {"role": "system", "content": SYSTEM_PROMPT},
    {"role": "user", "content": question + text}],
  "temperature": 0, "seed": 42, "max_tokens": 2000
}
```

What comes back

content	the response's text
model	the id that answered
usage	tokens in and out, billed
finish_reason	stop, or length

The model sees only the messages: not your file, not your folder, not your machine.

OpenRouter: one key, many models

20

free-model requests a minute

50

a day, under 10 credits bought

1,000

a day, 10 credits or more

Key	Model	Price per million tokens
text	nvidia/nemotron-3-super-120b-a12b:free	US$0.00 in, US$0.00 out
fallback	google/gemini-2.5-flash-lite	US$0.10 in, US$0.40 out
vision	google/gemini-2.5-flash-lite (day 13)	the same

Sources, read 8 October 2026: openrouter.ai/docs/api-reference/limits; openrouter.ai/api/v1/models

The key

get_secret looks in order

1. the environment
2. Colab's Secrets panel
3. my_bondmath\.env

Never

print the client (client.api_key shows it)

paste it in a cell, a prompt, a command

commit .env

```
!git -C "{work}" check-ignore -v .env cache/llm/
.gitignore:4:.env    .env
.gitignore:5:cache/    cache/llm/
```

The key cell prints only: OpenRouter key loaded: False. Mode: replay.

ask: one door for every request

1 System prompt first	quote exactly, give the section, NOT IN TEXT, refuse opinions
2 A fingerprint	request_key: SHA-256 of provider, model, messages, parameters
3 The cache	one JSON file per key; replay returns it or raises LookupError
4 Refusals	429 and 5xx waited and retried; others stop with a plain message

No header and no key in the fingerprint or the file. pytest at the end of the day: 59 passed.

The first request

The Committee decided to raise the target range for the federal funds rate by 1/4 percentage point to 3-3/4 to 4 percent, in support of the Federal Reserve's dual mandate.

```
Model nvidia/nemotron-3-super-120b-a12b:free, temperature 0, seed 42; saved response recorded 2026-10-08; replayed (no key loaded).
```

3-3/4 to 4 percent

in the statement, by day 1's regex

3-3/4 to 4 percent

in the saved response

yes

the same range

A number a model returns is checked against the document by code that does not involve the model.

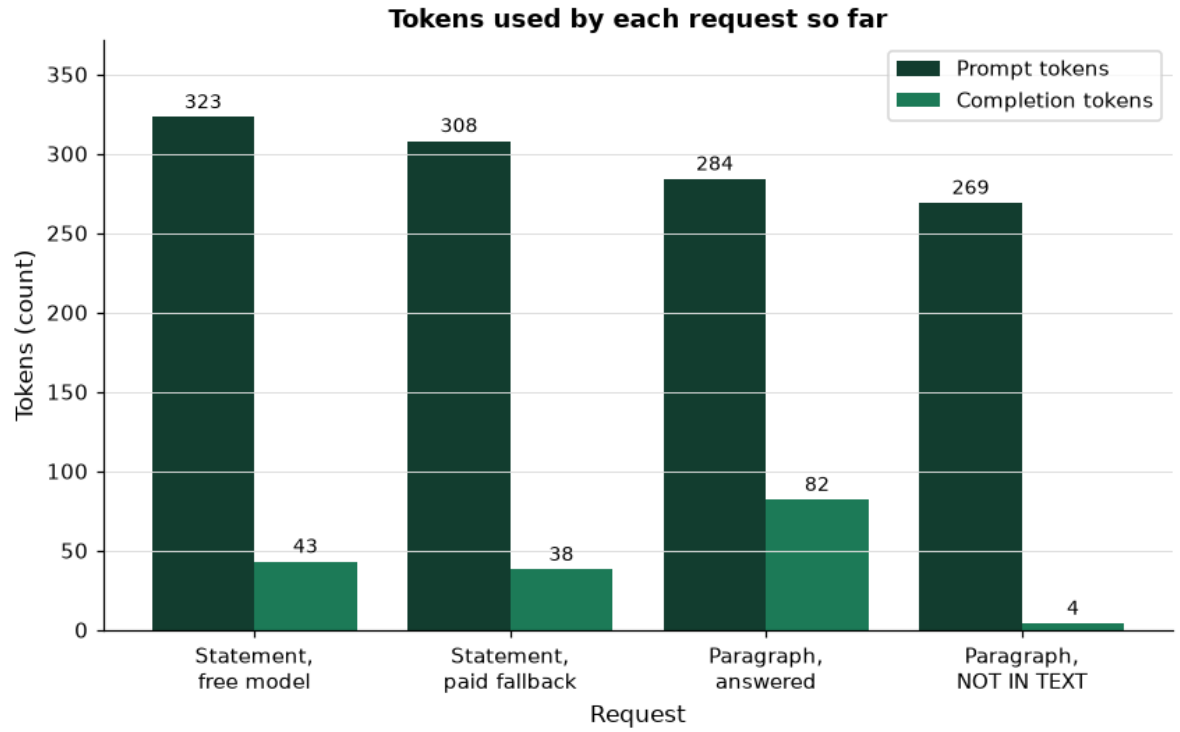
The system prompt on one paragraph

Question	Saved response
Share redeemable with Equity Offering proceeds, and the price?	40%, 104.750%: each in the paragraph
What is the interest rate on the Notes?	NOT IN TEXT

Section 3.07(b), 513 characters. The rate is in the indenture's title, not in the paragraph sent.

Source: Boyd Gaming Corporation, indenture for the 4.750% Senior Notes due 2031, Exhibit 4.1 to the 8-K filed 2021-06-08, accession 0001193125-21-185495, Section 3.07.

Tokens and cost



323 / 43

prompt / completion, statement, free model

308 / 38

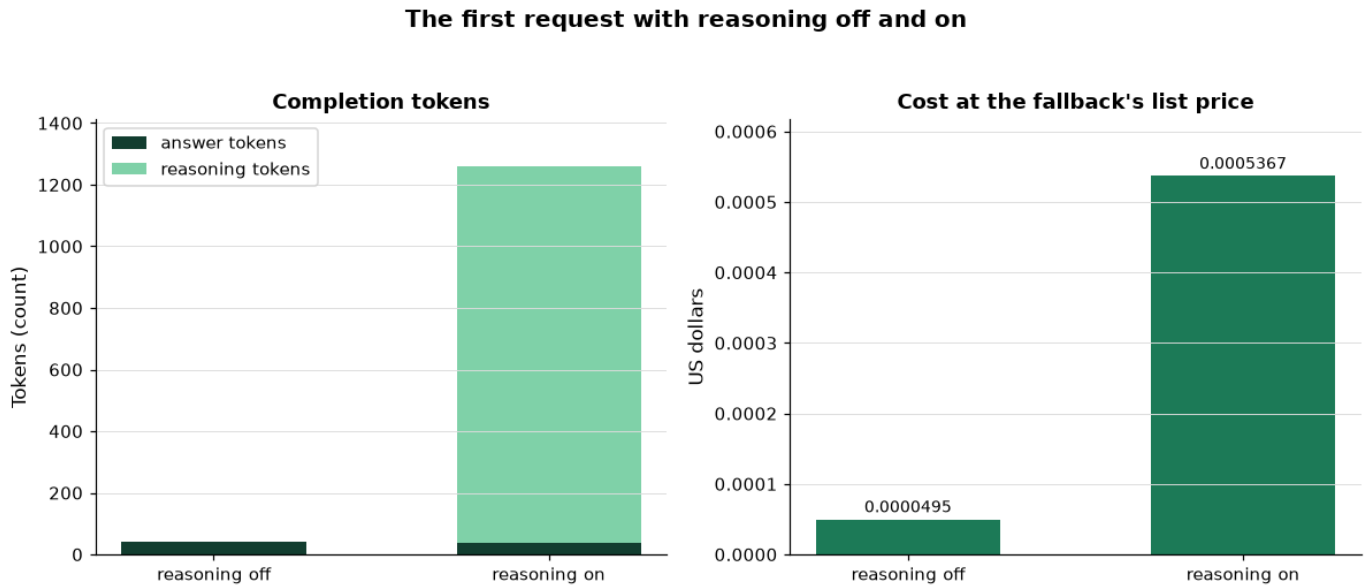
the fallback's own count of the same text

US$0.0000460

usage_cost, equal to the reported cost

```
cost = prompt_tokens x prompt price + completion_tokens x completion price   # US dollars
```

Reasoning switched off



43 / 1261

completion tokens, reasoning off / on; 1222 of the second are reasoning, never shown

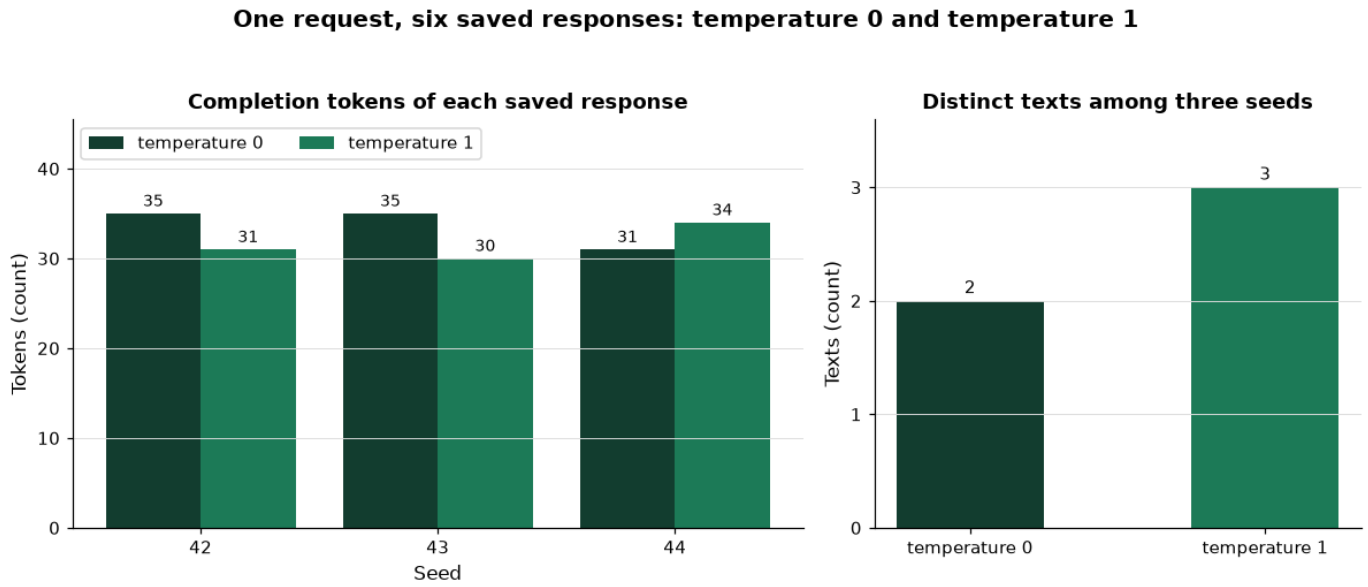
Same sentence quoted either way. The answer is in the text, so the course asks the model to find and quote; reasoning tokens are completion tokens, and a paid model bills them.

```
extra_body=textkit.NO_REASONING   # {"reasoning": {"enabled": false}}
```

How temperature and the seed work

The problem	A model writes one token at a time: how is the next one drawn?
The draw	Every possible next token has a probability, and one is drawn
Temperature	Scales the probabilities: 0 takes the likeliest nearly always, 1 follows the model's own
The seed	Asks for the same draws on a rerun: a request, not a promise
What comes out	Text that narrows toward one answer as the temperature falls, and does not repeat exactly

Temperature and seed



2

distinct texts of 3 at temperature 0; mean 33.7 tokens

3

distinct texts of 3 at temperature 1; mean 31.7 tokens

Temperature 0 and a seed narrow the spread. They do not promise identical text.

Replay, and when a request fails

```
cache/llm/0e28d64e7792....json   (14 saved responses: request, content, usage, recorded; no header, no key)
two words added, replay: LookupError: no saved response for request dd2ca18e0b82 in llm
```

```
a wrong model id (HTTP 400): The provider refused the request with HTTP 400
a refused key (HTTP 401): The provider refused the key: check OPENROUTER_API_KEY in your .env
no credit on a paid model (HTTP 402): The provider refused the request with HTTP 402
```

400 bad request: a wrong model id

401 the key

402 no credit

429 too many: a wait can clear a per-minute limit, not a daily cap

The OpenAI route

```
PROVIDERS = {"openrouter": {"base_url": "https://openrouter.ai/api/v1", "key_name": "OPENROUTER_API_KEY"},
             "openai":     {"base_url": "https://api.openai.com/v1",    "key_name": "OPENAI_API_KEY"}}
```

gpt-4.1-mini-2025-04-14

the model id OpenAI returned

None

provider: an OpenRouter field

The provider is part of the fingerprint: one saved response per route.

How the settings of a request work

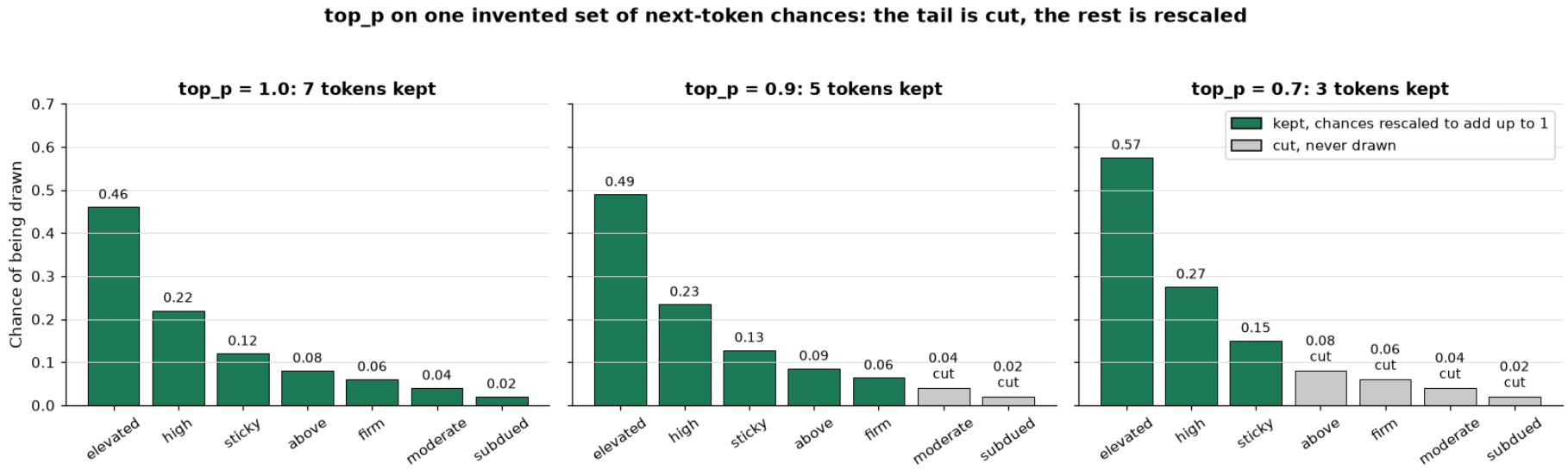
The problem	Which settings shape how the answer is drawn and delivered?
The draw	temperature rescales the chances, top_p cuts the tail, seed asks for the same draws
Length and form	max_tokens stops the answer; response_format asks for its shape
What you control	Only the settings you send; the package leaves out the rest
What comes out	The response of section 11.1

Every setting of the chat request

Setting	Default	The course sends	Group
messages	required	set in every request	the request
model	required	set in every request	the request
max_tokens	not sent	2000	length
response_format	not sent	None	shape of the answer
seed	not sent	42	randomness
temperature	not sent	0.0	randomness
extra_body	None	reasoning off (11.8)	the transport

41 settings, read from the installed package; the course sends 7, and no top_p. The rest are left out, and the provider uses its own defaults.

top_p drawn by hand




```
top_p 1.0: keeps 7 of 7 tokens; the largest chance is now 0.460
top_p 0.9: keeps 5 of 7 tokens; the largest chance is now 0.489
top_p 0.7: keeps 3 of 7 tokens; the largest chance is now 0.575
```

Sort, keep until the chances reach top_p, cut the rest, rescale. The chances are invented, not a model's output.

Four answers side by side

free model, reasoning off

43 tokens; same range: yes; cites a section: no

The Committee decided to raise the target range for the federal funds rate by 1/4 percentage point to 3-3/4 to 4 percent, in support of the Federal Reserve's dual mandate.

free model, reasoning on

1,261 tokens; same range: yes; cites a section: yes

"The Committee decided to raise the target range for the federal funds rate by 1/4 percentage point to 3-3/4 to 4 percent," - Section 1

paid fallback

38 tokens; same range: yes; cites a section: yes

"The Committee decided to raise the target range for the federal funds rate by 1/4 percentage point to 3-3/4 to 4 percent" (Section 1)

OpenAI

61 tokens; same range: yes; cites a section: yes

The statement says about the target range for the federal funds rate:
"The Committee decided to raise the target range for the federal funds rate by 1/4 percentage point to 3-3/4 to 4 percent, in support of the Federal Reserve's dual mandate." (Section 1)

4 of 4 give the statement's range; 3 cite a section, and a statement has none. A tidy answer is not a checked one.

Excel and Git

Excel side by side (tested in Excel)

`=B2*C2+B3*C3`	0.000046
`=SUMPRODUCT(B2:B3,C2:C3)`	0.000046
`=TEXT(D1,"0.0000000")`	0.0000460
`=B2*E2+B3*E3  (prices as text)`	0.000046
`=SUMPRODUCT(B2:B3,E2:E3)  (prices as text)`	0

```
!! __pycache__/
!! cache/
```

SUMPRODUCT treats text cells as zero. git status --ignored: cache/ marked !!, out of Git.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Units, conventions, and what the function must never do: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the helper prints every header, and Authorization holds the whole key.

Code that runs is not the same as code that is right.

What you can do now

- Say what a request holds and what comes back
- Keep the key in .env, out of Git, never printed
- Send through ask; check numbers with code
- Price tokens with usage_cost, free or paid
- Replay saved responses; read 400 to 429

Week 3, Day 2 Covenants 1: the anatomy of an indenture